`Lipa City Colleges
College of Computing and Technology Engineering
THESIS 1

Bubble Vision: An Automated Bubble Test Checker and Item Analysis System using K – Nearest Neighbors

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SUPER USER SIDE

1. **Authentication**
    - Username
    - Password
2. **Dashboard**
    - Total number of institutions
    - Total number of institution admin
    - Total number of teachers
3. **Institution**
    - Add, Edit, Delete Institution
        - Name,
        - Address,
        - School year**Institution admin**
    - Add, Edit, Delete institution admin
        - Username
        - First name
        - Last name
    - Default password: @INSADMIN2025
4. **Assign admin**
    - Add by filtering the existing admin and institution
    - Edit saved admin of that institution
    - Delete saved admin of that institution
5. **Logout**

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ADMIN SIDE

1. **Authentication**
    - Username
    - Password
2. **Dashboard**
    - Display Institution and school year
    - Welcome, {name}
    - Total number of users
    - Total number of students
    - Total number of grade level
    - Total number of subjects
3. **Grades**
    - Add Grade
    - Edit Grade
    - Delete Grade
4. **Section**
    - Add Section
    - Edit Section
    - Delete Section
    - Search Section
5. **Subject**
    - Add Subject
    - Edit Subject
    - Delete Subject
    - Search Subject
6. **Students**
    - Add, Edit, Delete Student
        - Grade
        - Section
        - First name
        - Middle initial

    - Last name
  - Import student via docs, txt, csv, pdf
  - Display list of students per section and grade
7. **Users / Teachers**
  - Add, Edit, Delete User/Teacher
    - First name
    - Last name
    - Username
    - Grade
    - Section
    - Subject
  - Default password: @Teacher2025
8. **Profile**
  - Display account details
    - Full name
    - Email
    - Username
    - Role
    - Date joined
    - Date and time of last login
  - Change password
    - Old password
    - New password
    - New password confirmation
9. **Logout**

USER/TEACHER SIDE

1. **Authentication**
    - Username
    - Password
2. **Dashboard**
    - Display Institution and school year
    - Welcome, {name}
    - Total number of students
    - Total number of subjects
    - Total number of scanned answer sheets
3. **Classes**
    - Add, Edit, Delete Class
        - Grade level
        - Section
        - Subject
        - Questionnaire
        - Answer key
4. **Review Answer key**
    - Display grade, section, subject
    - Change answer key
    - Change points
5. **Check bubble sheet via upload**
    - Student name
    - Chosen bubble sheet file
6. **Scan bubble sheet via phone**
    - Student name
    - Proceed scanning
7. **Scan result**
    - Display grade, section, subject
    - Student name
    - Score and its percentage

- Date and time scanned happen
- Bubble sheet image with mark detection
    - Red circle for wrong answer
    - Green circle for correct answer
    - Yellow circle for correct key but consider the student answer as invalid
- Per item results display
    - Item
    - Correct key
    - Detected student answer
    - Points of that item
    - Status of the answer
        - Correct – same answer to the key
        - Wrong – different answer to the key
        - Invalid – same answer to the key but the filled is invalid either not fully shaded, has crossed, or 2 or more was shaded in that item
        - Blank – no answer

8. Scan History

- List of students that the paper already scanned
    - Student name
    - Score
    - Date and time it was scanned
    - Delete student canned result

9. Item analysis

- Display grade, section, subject
- Total number of students
- Total items
- Item analysis result
    - Item
    - Correct key
    - Number of students correct on that item
    - Percentage equivalent of the student correct per item

- **Range of Difficulty index** | **Difficulty indicator** | **Action**
  - 0 – 0.25 (0% - 25%) | Difficult | Revise or discard
  - 0. 26 – 0.75 (26% - 75%) | Right difficult | Retain
  - 0.76 – above (76% - above) | Easy | Revise or discard

10. Reports

- Scanned result per student
- Scanned result of the class
- Students score
- Item analysis result

11. Logout